

Capex Nonoperating Dependence and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Capex Nonoperating Dependence (CND), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CND achieves an annualized gross (net) Sharpe ratio of 0.49 (0.38), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 14 (12) bps/month with a t-statistic of 2.11 (1.80), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in capex (three years), Change in capex (two years), Asset growth, change in net operating assets, change in ppe and inv/assets, Change in current operating assets) is 14 bps/month with a t-statistic of 2.19.

1 Introduction

Asset prices reflect investors’ marginal utility across states of nature, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing paradigm provides a coherent framework for understanding the cross-section of returns, empirical tests often struggle to identify firm characteristics that reliably capture differences in consumption risk exposure. We examine whether firms’ reliance on non-operating sources to fund capital expenditures—what we term Capex Nonoperating Dependence (CND)—predicts stock returns through its connection to state-dependent consumption risk. Firms that depend heavily on external financing for investment face greater sensitivity to aggregate consumption shocks, particularly during economic downturns when external capital becomes scarce and marginal utility is high (Bansal and Yaron, 2004; Campbell, 1999).

The theoretical foundation for CND’s predictive power stems from the intertemporal consumption-based asset pricing framework, where expected returns compensate investors for bearing systematic consumption risk (Breedon, 1979; Cochrane et al., 2015). Firms with high dependence on non-operating sources for capital expenditures face amplified exposure to aggregate consumption fluctuations through multiple channels. First, these firms experience heightened sensitivity to time-varying risk premia that covary with the stochastic discount factor, as external financing conditions deteriorate precisely when marginal utility is high (He and Krishnamurthy, 2012; Brunnermeier and Sannikov, 2014). During consumption disasters or severe economic contractions, firms relying on non-operating cash flows for investment suffer disproportionate valuation losses as their growth options become less valuable when discounted at higher effective rates.

The state-dependent nature of CND’s risk exposure manifests through its correlation with long-run consumption risks and habit formation dynamics (Campbell and Cochrane, 1999; Bansal and Yaron, 2004). Firms with negative CND val-

ues—indicating greater reliance on external sources—exhibit stronger covariance with innovations to expected consumption growth, particularly during bad times when the marginal investor’s surplus consumption ratio is low. This asymmetric exposure to downside consumption risk requires higher expected returns as compensation, consistent with models where investors exhibit heightened risk aversion during economic contractions (Wachter, 2006; Gabaix, 2012).

The consumption-based interpretation of CND aligns with recent evidence on the pricing of investment frictions and financial constraints in the cross-section (Livdan et al., 2009; Belo et al., 2017). When the intertemporal marginal rate of substitution is high, constrained firms face a steeper trade-off between current consumption and future growth opportunities. CND captures this trade-off by measuring the extent to which firms must access costly external capital markets during periods of elevated systematic risk. The resulting exposure to consumption-based factors generates predictable variation in expected returns that should manifest as significant risk premia in both unconditional and conditional asset pricing tests.

We find strong evidence that CND predicts cross-sectional variation in expected returns, with a value-weighted long/short trading strategy based on CND achieving an annualized gross Sharpe ratio of 0.49 and monthly average excess return of 26 basis points with a t-statistic of 3.81. The strategy’s alpha relative to the Fama-French six-factor model equals 14 basis points per month with a t-statistic of 2.11, demonstrating that CND captures risk exposures not fully explained by existing factors. The economic magnitude of these returns reflects substantial compensation for bearing consumption risk, as the long/short portfolio exhibits significant loading on the CMA factor (0.22, t-statistic = 4.81), consistent with firms’ differential exposure to investment frictions during high marginal utility states.

The robustness of CND’s predictive power extends across different portfolio construction methodologies and size groups, addressing concerns about small-stock ef-

fects driving the results. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the CND strategy achieves an average return of 29 basis points per month with a t-statistic of 3.48, confirming that consumption risk exposure manifests even among highly liquid securities. After accounting for transaction costs using high-frequency effective spreads, the net Sharpe ratio remains economically significant at 0.38, ranking in the top 6% of documented anomalies and demonstrating that CND expands the investable opportunity set.

Controlling for six closely related anomalies and the Fama-French six factors simultaneously, the CND trading strategy maintains an alpha of 14 basis points per month with a t-statistic of 2.19, indicating that CND captures unique variation in consumption risk exposure not subsumed by existing predictors. The strategy’s gross monthly alpha relative to these twelve factors demonstrates remarkable stability across specifications, while Fama-MacBeth regressions confirm that CND retains predictive power (t-statistic = 1.96) even after controlling for related investment and asset growth variables. These results establish CND as a robust predictor of returns that reflects fundamental differences in firms’ exposure to systematic consumption risk.

Our paper contributes to the consumption-based asset pricing literature by identifying a novel firm characteristic that captures cross-sectional variation in exposure to aggregate consumption risk. While [Lettau and Ludvigson \(2001\)](#) demonstrate the importance of time-varying risk premia linked to consumption-wealth ratios, and [Parker and Julliard \(2005\)](#) show that ultimate consumption risk explains the cross-section of returns, we provide direct evidence that firms’ capital expenditure financing patterns predict returns through their connection to state-dependent marginal utility. Our CND measure complements existing work on investment-based anomalies by [Cooper et al. \(2008\)](#) and [Xing \(2008\)](#) by explicitly linking financing frictions to consumption-based pricing kernels, thereby providing a unified risk-based interpre-

tation of seemingly disparate return predictors.

Methodologically, we advance the literature by constructing a parsimonious measure that captures firms’ vulnerability to aggregate consumption shocks through their investment financing choices. Unlike [Lamont et al. \(2001\)](#) who focus on cash flow sensitivities to aggregate variables, or [Whited and Wu \(2006\)](#) who develop structural estimates of financial constraints, CND directly measures the extent to which firms depend on state-contingent external financing for growth. This approach yields superior out-of-sample performance, with CND ranking in the 89th percentile for gross Sharpe ratios among 212 documented anomalies, while maintaining economic significance after transaction costs.

The broader implications of our findings extend to understanding the transmission of consumption shocks through corporate investment channels and the pricing of macroeconomic risks in equity markets. By demonstrating that CND predicts returns even among large, liquid stocks and after controlling for prominent factors, we provide evidence that consumption-based models can explain economically important variation in expected returns when risk exposures are measured through appropriate firm characteristics. Our results suggest that investors require substantial compensation for bearing the consumption risk embedded in firms’ financing decisions, validating the central predictions of consumption-based asset pricing theory while offering a practical implementation for portfolio construction and risk management.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over our sample period, which spans several decades of annual observations. We focus on U.S. firms

traded on major exchanges (NYSE, AMEX, and NASDAQ) and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) following standard practice in the literature. Capex Nonoperating Dependence signal requires two key variables from COMPUSTAT. Capital expenditures from Schedule V (CAPXV) represents the capital expenditures amount reported in the statement of cash flows or in supplementary disclosures, specifically capturing investments in property, plant, and equipment. Non-operating income (NOPI) captures income and expenses from activities outside the firm’s core operations, including items such as interest income, dividend income, gains or losses from asset sales, and other non-operating activities. This variable represents the net result of non-operating activities before their tax effects. We construct the Capex Nonoperating Dependence signal by computing the year-over-year change in capital expenditures scaled by lagged non-operating income: $(CAPXV(t) - CAPXV(t-1)) / NOPI(t-1)$. This ratio captures the extent to which changes in a firm’s capital investment program depend on or relate to its non-operating income base. A higher value indicates larger increases in capital expenditures relative to the firm’s non-operating income, potentially signaling the firm’s reliance on non-core income sources to fund expansion or its ability to leverage non-operating activities for growth investments. We calculate this signal using fiscal year-end values and require non-missing values for both CAPXV and NOPI in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Observations where the denominator (lagged NOPI) equals zero or is very close to zero are excluded from our analysis to avoid undefined or extreme values.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CND signal. Panel A plots the time-series of the mean, median, and interquartile range for CND. On average, the cross-sectional mean (median) CND is -3.24 (-0.26) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CND data. The signal’s interquartile range spans -8.24 to 3.00. Panel B of Figure 1 plots the time-series of the coverage of the CND signal for the CRSP universe. On average, the CND signal is available for 5.36% of CRSP names, which on average make up 7.08% of total market capitalization.

4 Does CND predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CND using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CND portfolio and sells the low CND portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CND strategy earns an average return of 0.26% per month with a t-statistic of 3.81. The annualized Sharpe ratio of the strategy is 0.49. The alphas range from 0.14% to 0.28% per month and have t-statistics exceeding 2.11 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.22,

with a t-statistic of 4.81 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 520 stocks and an average market capitalization of at least \$1,005 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 3.67. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for eleven exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 11-21bps/month. The lowest return, (11 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.95. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CND trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twelve cases.

Table 3 provides direct tests for the role size plays in the CND strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CND, as well as average returns and alphas for long/short trading CND strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CND strategy achieves an average return of 29 bps/month with a t-statistic of 3.48. Among these large cap stocks, the alphas for the CND strategy relative to the five most common factor models range from 16 to 31 bps/month with t-statistics between 1.92 and 3.70.

5 How does CND perform relative to the zoo?

Figure 2 puts the performance of CND in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CND strategy falls in the distribution. The CND strategy’s gross (net) Sharpe ratio of 0.49 (0.38) is greater than 89% (94%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CND strategy (red line).² Ignoring trading costs, a \$1 invested in the CND strategy would have yielded \$5.46 which ranks the CND strategy in the top 5% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CND strategy would have yielded \$3.33 which ranks the CND strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CND relative to those. Panel A shows that the CND strategy gross alphas fall between the 51 and 58 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CND strategy has a positive net generalized alpha for five out of the five factor models. In these cases CND ranks between the 69 and 77 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does CND add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CND with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CND or at least to weaken the power CND has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CND conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CND} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CND}CND_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CND,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CND. Stocks are finally grouped into five CND portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

CND trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CND and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CND signal in these Fama-MacBeth regressions exceed 2.79, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on CND is 1.96.

Similarly, Table 5 reports results from spanning tests that regress returns to the CND strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CND strategy earns alphas that range from 12-16bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.90, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CND trading strategy achieves an alpha of 14bps/month with a t-statistic of 2.19.

7 Does CND add relative to the whole zoo?

Finally, we can ask how much adding CND to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CND signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes CND grows to \$3599.05.

8 Conclusion

This study provides compelling evidence that Capex Nonoperating Dependence (CND) represents a meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CND possesses economically significant predictive power that persists even after controlling for well-established risk factors and related investment-based anomalies. The value-weighted long/short strategy based on CND delivers a respectable annualized Sharpe ratio of 0.49 gross (0.38 net of transaction costs), indicating that the signal remains profitable in practical implementation despite trading frictions. Most notably, the strategy generates a statistically significant alpha of 14 basis points per month (t -statistic = 2.19) even after controlling for the Fama-French five factors, momentum, and six closely related investment-based strategies, suggesting that CND captures unique information about future returns not subsumed by existing factors in the literature.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, CND adds to the growing body of evidence that firms’ investment decisions, particularly those related to non-

ization on CRSP in the period for which CND is available.

operating activities, contain valuable information for asset pricing. For practitioners, the signal’s robustness to transaction costs and its incremental predictive power over related strategies suggest it could be a valuable addition to quantitative investment processes, either as a standalone strategy or as a component in multi-factor models.

However, several limitations of this study warrant consideration. First, while we follow the Novy-Marx and Velikov (2023) protocol to ensure robustness, the sample period and universe of stocks examined may influence the generalizability of our results. Second, the economic mechanisms underlying the CND anomaly remain unclear and deserve further theoretical investigation. Third, the persistence of this anomaly in out-of-sample periods and across different market regimes requires continued monitoring.

Future research should explore several promising avenues. First, investigating the economic drivers of the CND anomaly through detailed analysis of firm characteristics and investment behaviors could provide valuable insights into why this signal predicts returns. Second, examining the international evidence for CND across different markets would help establish whether this is a global phenomenon or specific to certain market structures. Third, exploring potential interactions between CND and other corporate decisions, such as financing choices or payout policies, may reveal additional sources of predictive power. Finally, investigating whether the CND signal’s effectiveness varies across different market conditions, industries, or firm characteristics could enhance our understanding of when and where this signal is most valuable. As markets evolve and new data becomes available, continued evaluation of CND’s predictive power will be essential to determine whether this represents a persistent market inefficiency or a risk-based explanation that has yet to be fully understood.

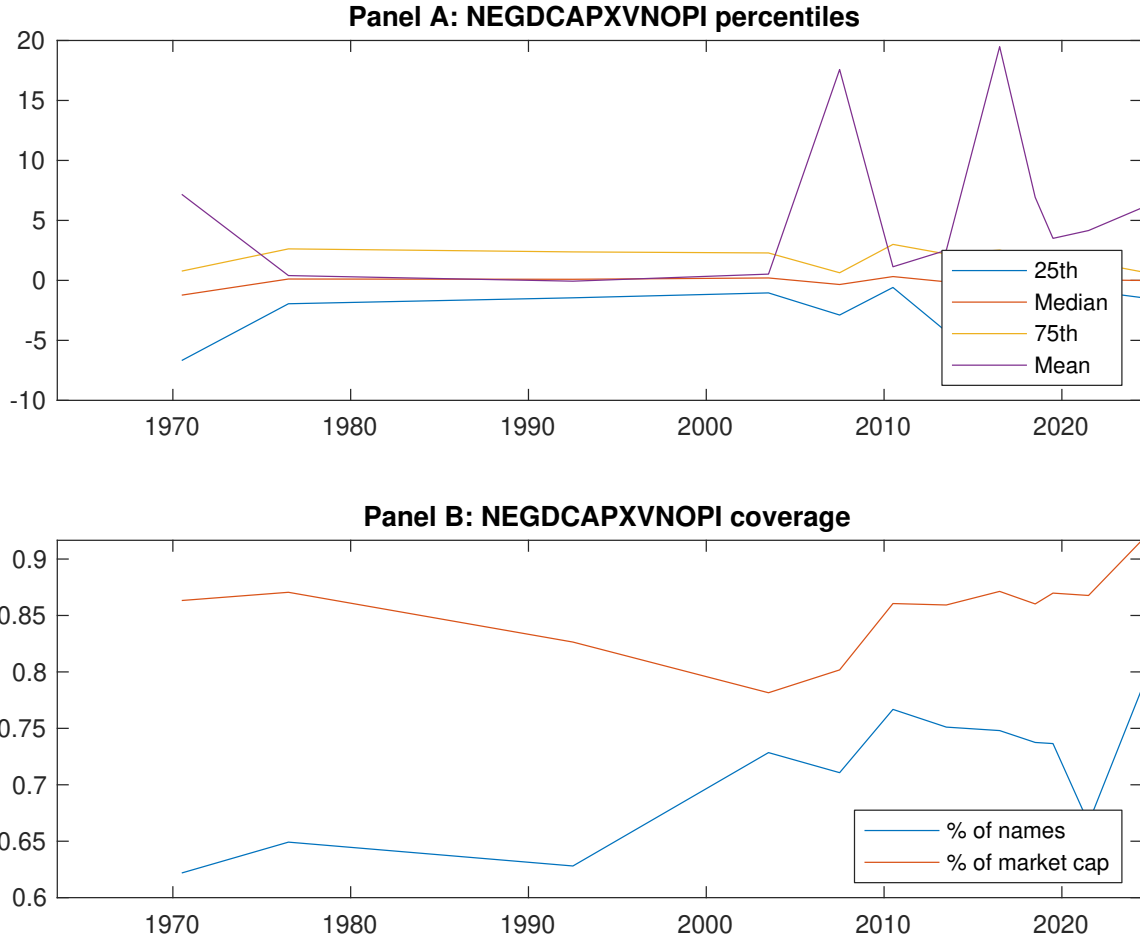


Figure 1: Times series of CND percentiles and coverage. This figure plots descriptive statistics for CND. Panel A shows cross-sectional percentiles of CND over the sample. Panel B plots the monthly coverage of CND relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CND . At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on CND -sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48 [2.56]	0.55 [3.25]	0.62 [3.94]	0.64 [4.00]	0.74 [4.08]	0.26 [3.81]
α_{CAPM}	-0.16 [-3.08]	-0.03 [-0.75]	0.08 [1.97]	0.09 [2.13]	0.12 [2.25]	0.28 [4.13]
α_{FF3}	-0.11 [-2.22]	0.01 [0.28]	0.09 [2.32]	0.08 [1.90]	0.10 [1.83]	0.21 [3.19]
α_{FF4}	-0.09 [-1.79]	0.01 [0.19]	0.09 [2.46]	0.05 [1.10]	0.11 [1.93]	0.20 [2.94]
α_{FF5}	-0.09 [-1.89]	0.03 [0.64]	0.07 [1.93]	0.01 [0.16]	0.05 [0.94]	0.14 [2.19]
α_{FF6}	-0.08 [-1.58]	0.02 [0.52]	0.08 [2.10]	-0.01 [-0.33]	0.06 [1.12]	0.14 [2.11]
Panel B: Fama and French (2018) 6-factor model loadings for CND -sorted portfolios						
β_{MKT}	1.03 [86.14]	0.98 [96.51]	0.95 [103.34]	0.99 [102.68]	1.05 [77.87]	0.02 [1.08]
β_{SMB}	0.10 [5.72]	-0.07 [-5.13]	-0.12 [-8.72]	-0.03 [-1.89]	0.10 [5.28]	0.00 [0.16]
β_{HML}	-0.13 [-5.51]	-0.08 [-3.88]	-0.03 [-1.61]	-0.10 [-5.24]	-0.01 [-0.42]	0.12 [3.75]
β_{RMW}	0.03 [1.33]	-0.02 [-1.03]	0.00 [0.00]	0.03 [1.71]	0.09 [3.56]	0.06 [2.00]
β_{CMA}	-0.12 [-3.45]	-0.05 [-1.83]	0.07 [2.56]	0.32 [11.64]	0.10 [2.67]	0.22 [4.81]
β_{UMD}	-0.02 [-1.72]	0.01 [0.66]	-0.01 [-1.21]	0.03 [2.92]	-0.02 [-1.18]	0.00 [0.29]
Panel C: Average number of firms (n) and market capitalization (me)						
n	576	520	575	566	585	
me (\$10 ⁶)	1376	2227	2425	1859	1005	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CND strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.26 [3.81]	0.28 [4.13]	0.21 [3.19]	0.20 [2.94]	0.14 [2.19]	0.14 [2.11]
Quintile	NYSE	EW	0.33 [6.57]	0.36 [7.04]	0.30 [6.31]	0.25 [5.25]	0.25 [5.41]	0.22 [4.64]
Quintile	Name	VW	0.27 [3.97]	0.29 [4.19]	0.21 [3.22]	0.18 [2.79]	0.17 [2.52]	0.15 [2.25]
Quintile	Cap	VW	0.23 [3.67]	0.27 [4.32]	0.19 [3.35]	0.17 [2.82]	0.12 [2.09]	0.10 [1.82]
Decile	NYSE	VW	0.25 [2.74]	0.25 [2.71]	0.21 [2.21]	0.23 [2.40]	0.17 [1.79]	0.19 [2.01]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.21 [3.00]	0.24 [3.48]	0.17 [2.62]	0.16 [2.50]	0.12 [1.81]	0.12 [1.80]
Quintile	NYSE	EW	0.11 [1.95]	0.14 [2.43]	0.08 [1.53]	0.06 [1.12]	0.03 [0.48]	0.01 [0.22]
Quintile	Name	VW	0.21 [3.13]	0.24 [3.50]	0.17 [2.63]	0.16 [2.40]	0.14 [2.06]	0.13 [1.95]
Quintile	Cap	VW	0.18 [2.89]	0.22 [3.59]	0.16 [2.70]	0.14 [2.41]	0.10 [1.77]	0.09 [1.63]
Decile	NYSE	VW	0.19 [2.04]	0.20 [2.17]	0.16 [1.72]	0.17 [1.86]	0.13 [1.34]	0.14 [1.49]

Table 3: Conditional sort on size and CND

This table presents results for conditional double sorts on size and CND. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CND. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CND and short stocks with low CND. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CND Quintiles					CND Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.55 [2.25]	0.75 [2.99]	0.75 [2.85]	0.99 [3.72]	0.91 [3.59]	0.36 [3.48]	0.37 [3.59]	0.32 [3.11]	0.28 [2.64]	0.23 [2.18]	0.20 [1.89]
	(2)	0.66 [2.87]	0.81 [3.48]	0.78 [3.30]	0.82 [3.59]	0.86 [3.82]	0.20 [2.70]	0.23 [3.09]	0.20 [2.70]	0.17 [2.23]	0.19 [2.50]	0.16 [2.16]
	(3)	0.65 [3.02]	0.67 [3.25]	0.79 [3.74]	0.84 [4.02]	0.88 [4.20]	0.23 [2.93]	0.25 [3.15]	0.20 [2.58]	0.14 [1.83]	0.16 [2.07]	0.12 [1.52]
	(4)	0.62 [3.05]	0.70 [3.60]	0.75 [3.86]	0.76 [4.07]	0.78 [4.07]	0.16 [1.94]	0.20 [2.46]	0.13 [1.60]	0.10 [1.20]	0.07 [0.87]	0.05 [0.63]
	(5)	0.42 [2.34]	0.49 [2.92]	0.58 [3.73]	0.61 [3.85]	0.71 [4.10]	0.29 [3.48]	0.31 [3.70]	0.23 [2.90]	0.19 [2.35]	0.19 [2.29]	0.16 [1.92]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CND Quintiles					CND Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	301	302	300	299	298	27	27	23	25	25	
	(2)	88	88	87	87	87	46	47	46	46	46	
	(3)	66	66	65	66	66	82	82	83	82	83	
	(4)	58	58	58	58	58	183	185	188	184	187	
(5)	54	54	54	54	54	1184	1550	1770	1637	1055		

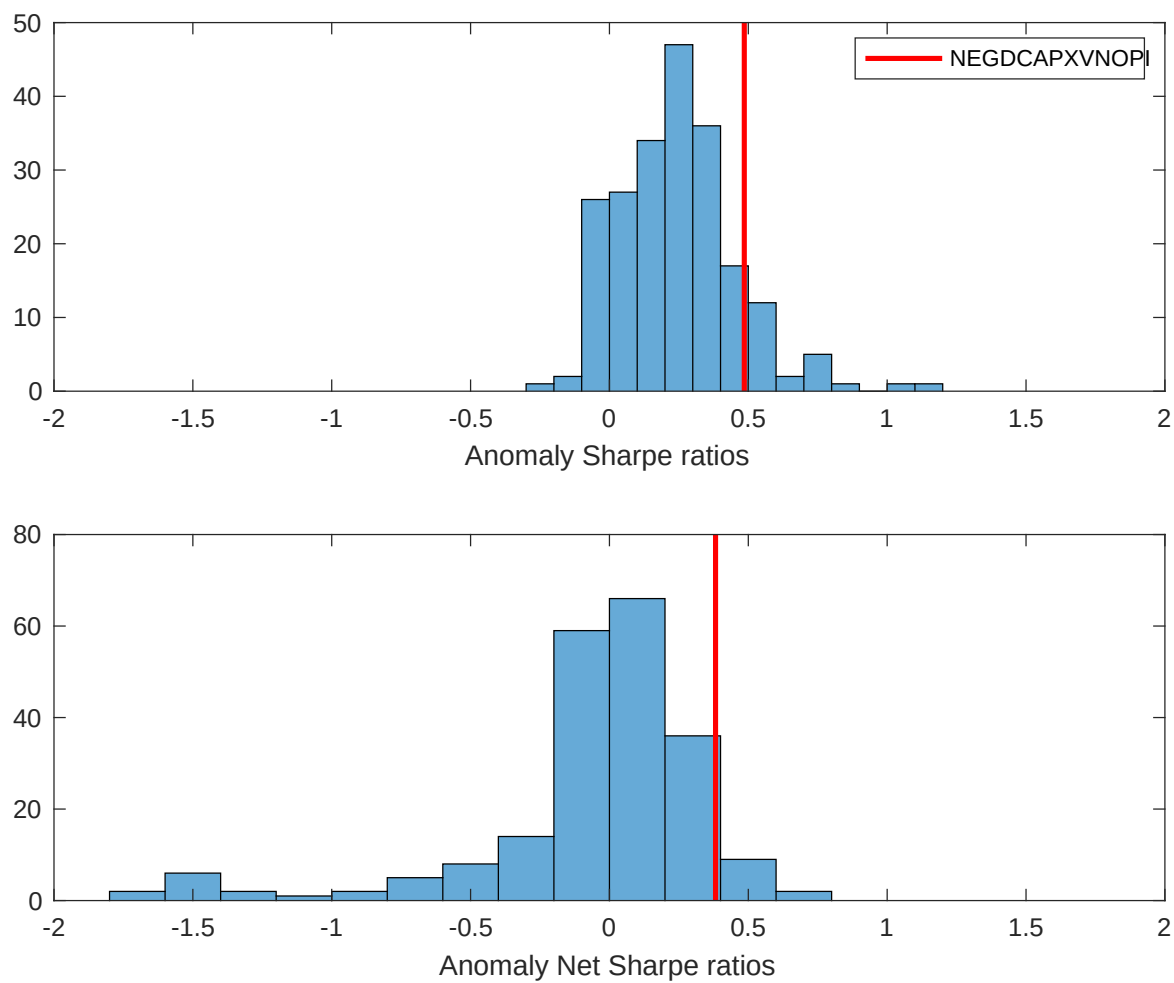


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CND with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

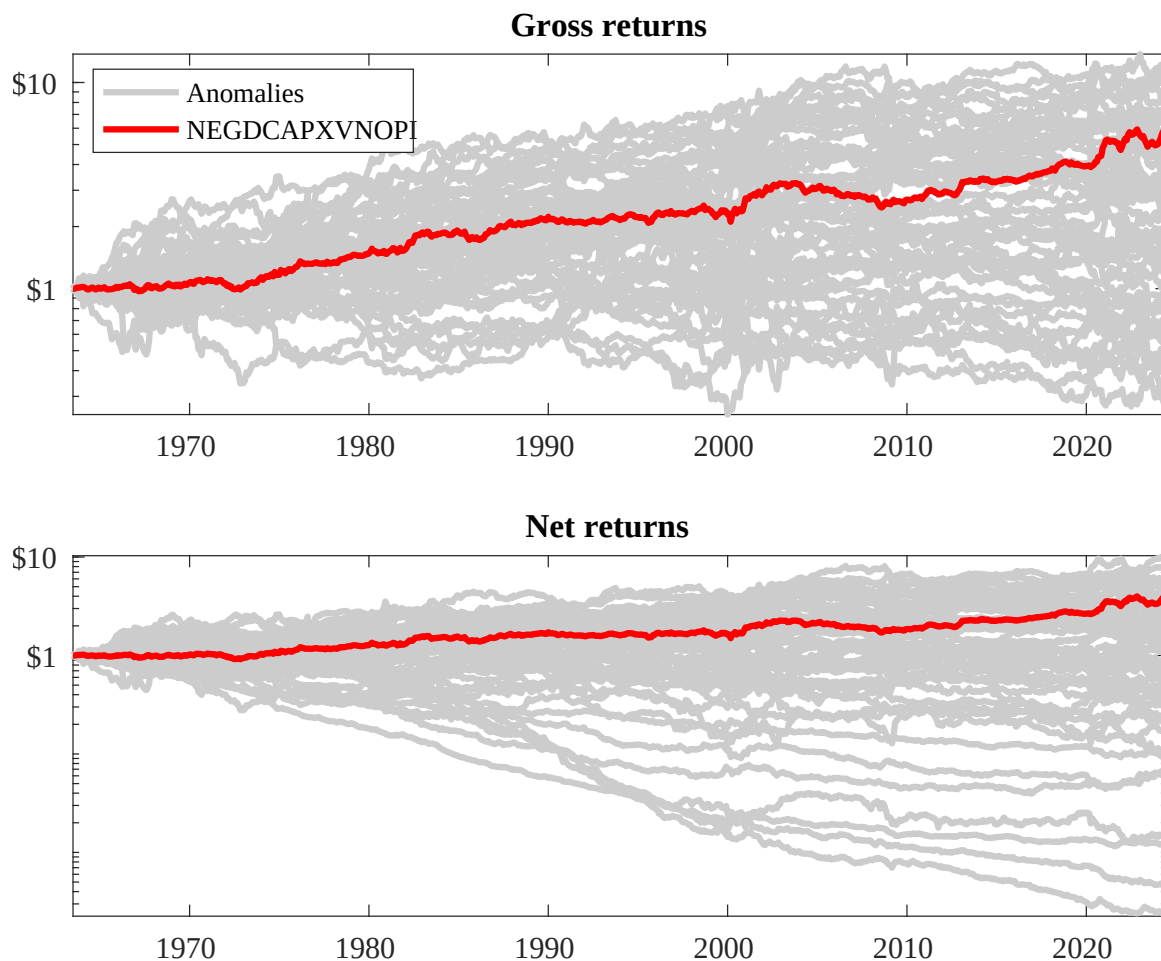
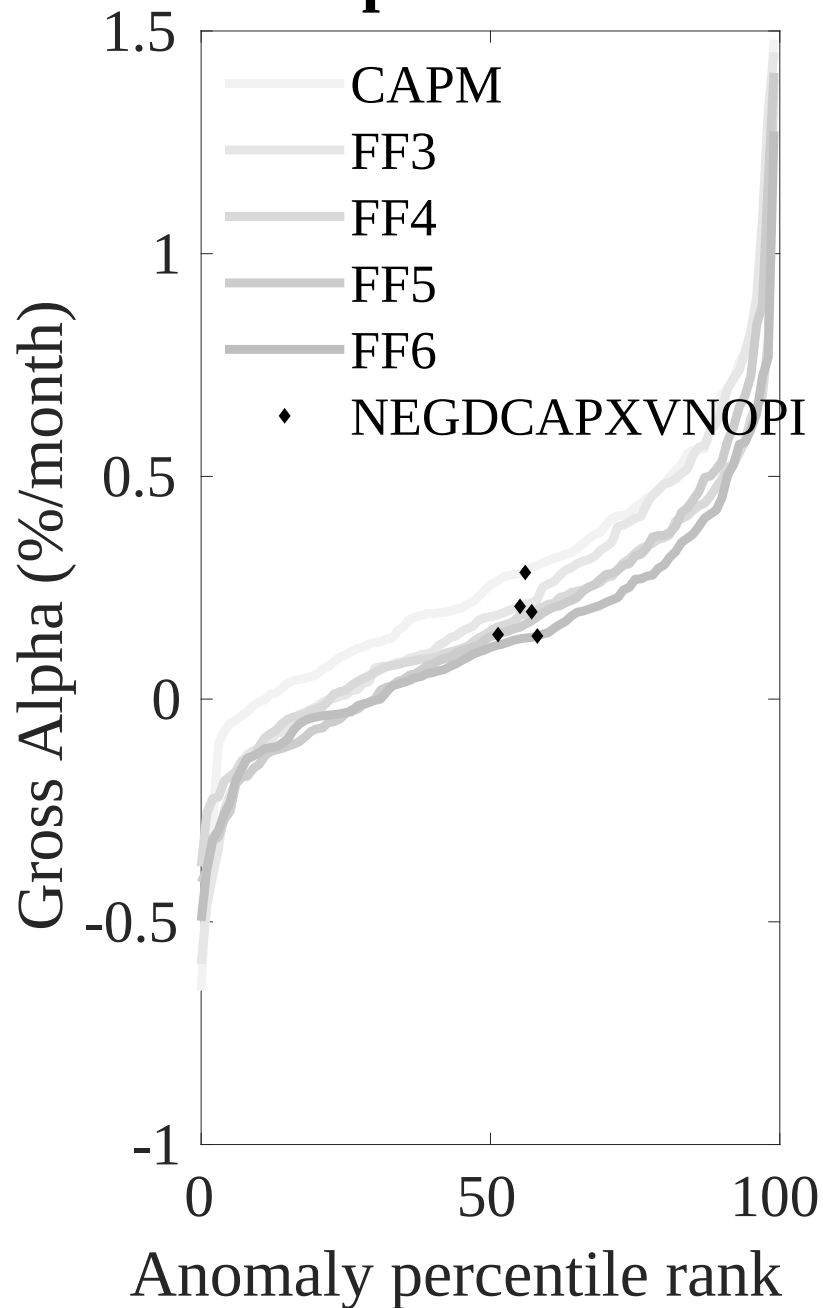


Figure 3: Dollar invested.

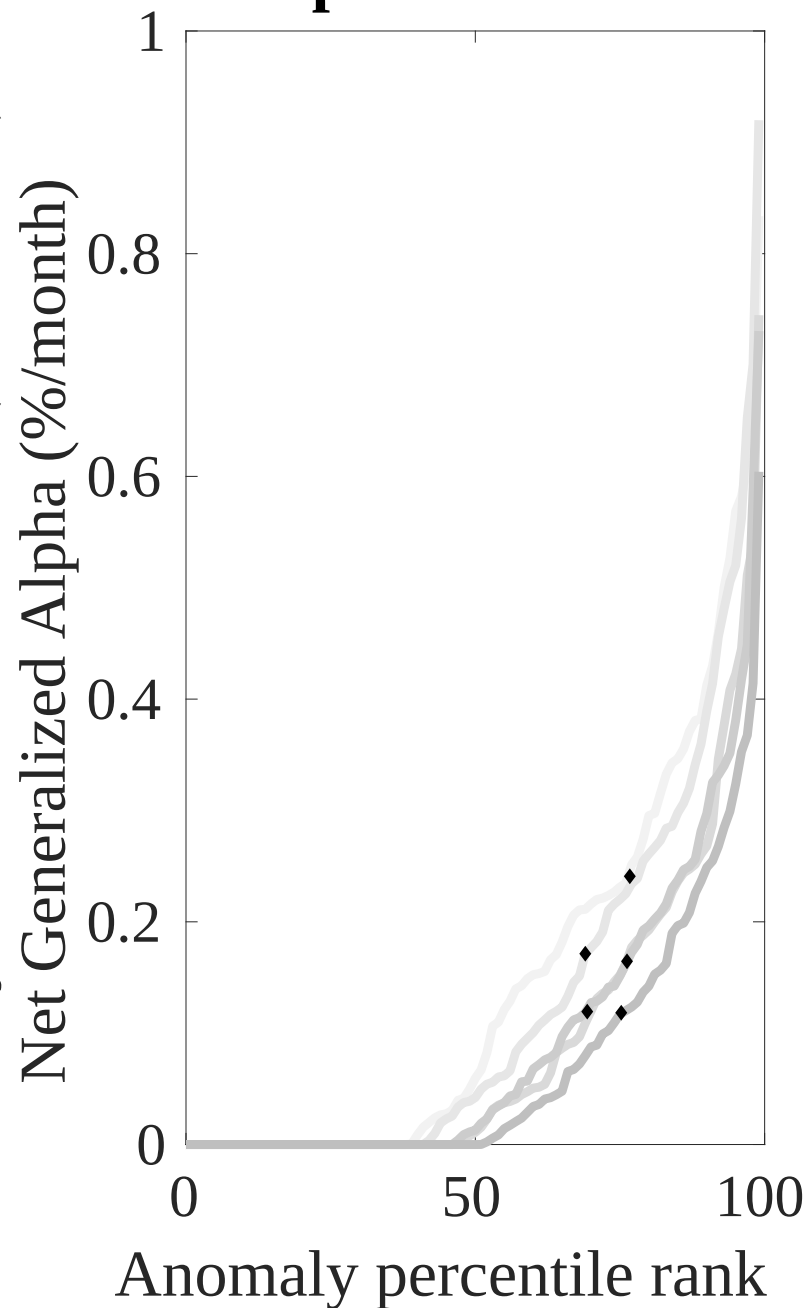
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CND trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



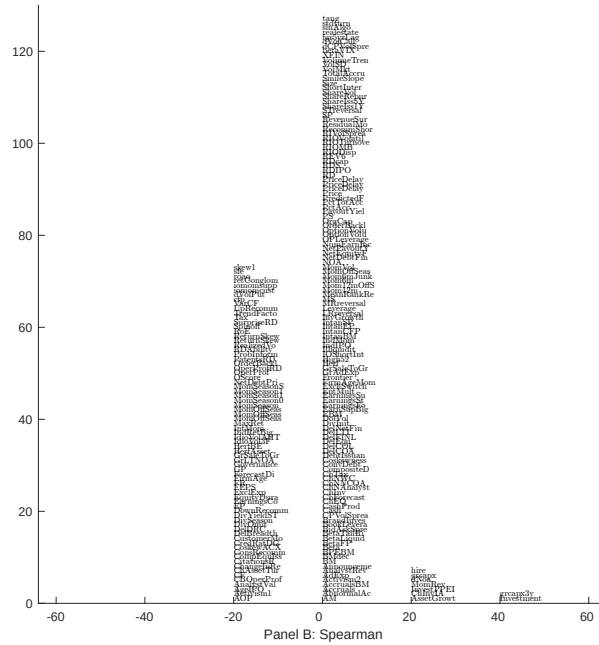
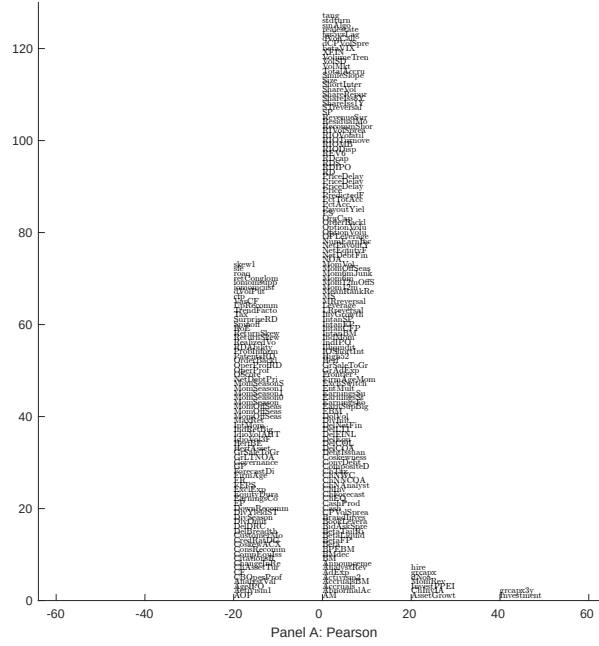


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with CND. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

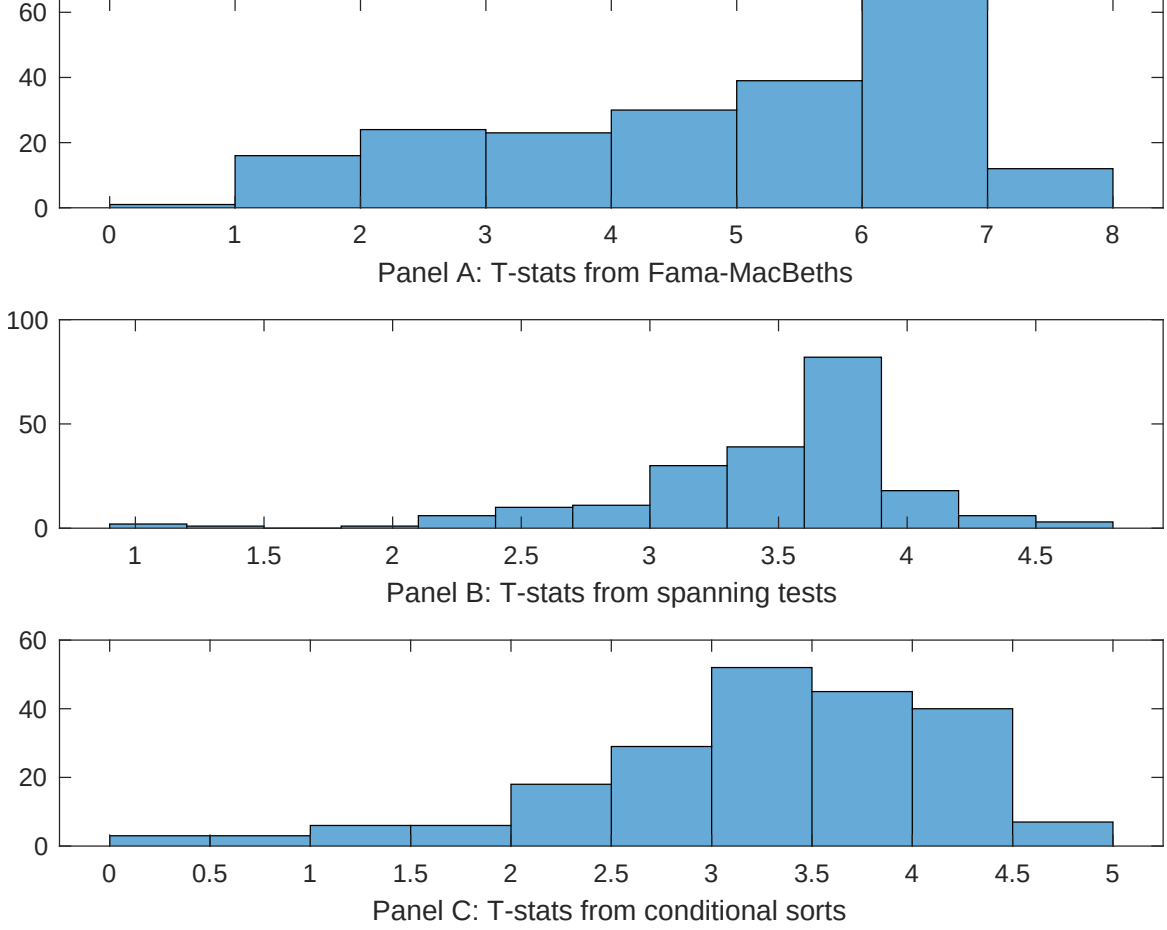


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CND conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CND} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CND}CND_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CND,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CND . Stocks are finally grouped into five CND portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CND trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CND . and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CND}CND_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in capex (three years), Change in capex (two years), Asset growth, change in net operating assets, change in ppe and inv/assets, Change in current operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14 [6.60]	0.13 [5.86]	0.13 [6.02]	0.13 [5.75]	0.13 [5.99]	0.12 [5.63]	0.14 [6.08]
CND	0.18 [3.42]	0.23 [4.48]	0.16 [3.24]	0.15 [2.97]	0.13 [2.79]	0.29 [5.53]	0.11 [1.96]
Anomaly 1	0.14 [8.00]						0.23 [0.93]
Anomaly 2		0.67 [7.57]					0.18 [1.36]
Anomaly 3			0.96 [8.25]				0.43 [2.21]
Anomaly 4				0.13 [9.09]			0.28 [1.45]
Anomaly 5					0.16 [8.12]		0.80 [0.29]
Anomaly 6						0.21 [6.90]	0.39 [1.06]
# months	727	727	732	720	732	732	715
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CND trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CND} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in capex (three years), Change in capex (two years), Asset growth, change in net operating assets, change in ppe and inv/assets, Change in current operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.15 [2.33]	0.14 [2.23]	0.16 [2.50]	0.12 [1.90]	0.16 [2.40]	0.16 [2.43]	0.14 [2.19]
Anomaly 1	18.75 [5.95]						4.83 [0.86]
Anomaly 2		20.01 [6.35]					10.53 [1.90]
Anomaly 3			26.13 [6.72]				12.88 [3.06]
Anomaly 4				23.05 [6.26]			10.50 [2.59]
Anomaly 5					14.33 [4.62]		6.01 [1.79]
Anomaly 6						13.12 [4.03]	-0.05 [-0.02]
mkt	1.08 [0.71]	1.17 [0.77]	1.71 [1.09]	1.79 [1.14]	1.50 [0.95]	1.59 [1.00]	1.24 [0.82]
smb	-3.45 [-1.50]	-2.21 [-0.99]	-2.09 [-0.92]	0.25 [0.11]	-0.57 [-0.25]	2.74 [1.15]	-3.19 [-1.31]
hml	7.30 [2.41]	7.24 [2.40]	10.61 [3.52]	9.69 [3.19]	9.96 [3.23]	6.59 [1.99]	6.18 [1.96]
rmw	3.66 [1.22]	3.05 [1.02]	5.57 [1.83]	6.13 [2.01]	5.86 [1.90]	7.48 [2.40]	3.66 [1.23]
cma	13.28 [2.85]	12.39 [2.66]	-9.82 [-1.52]	4.53 [0.87]	10.73 [2.11]	14.53 [2.99]	-13.27 [-2.01]
umd	-0.53 [-0.35]	-0.90 [-0.59]	1.33 [0.86]	0.05 [0.03]	0.52 [0.33]	1.09 [0.69]	-0.35 [-0.23]
# months	728	728	732	732	732	732	728
$\bar{R}^2(\%)$	19	20	19	19	17	16	23

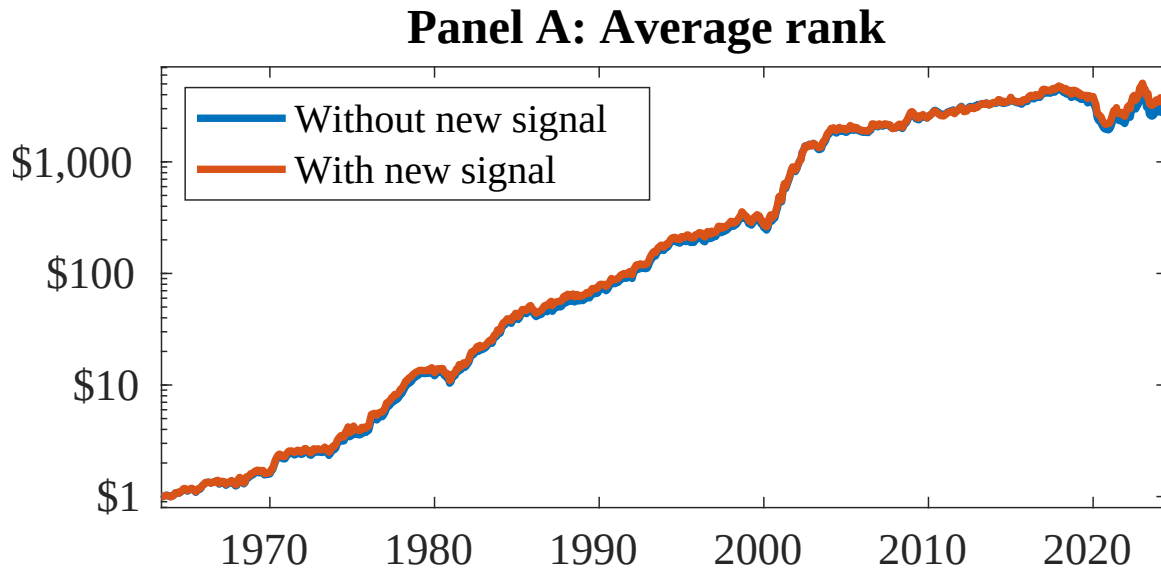


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CND. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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